Coexistence

Today's Agenda:

- Quiz
- Coexistence

Paradox of the plankton

Vol. XCV, No. 882 The American Naturalist May–June, 1961

THE PARADOX OF THE PLANKTON*

G. E. HUTCHINSON
Osborn Zoological Laboratory, New Haven, Connecticut

- Species need different niches to coexist
- How can so many species of plankton coexist?
  - All need the same resources
  - Well-mixed, homogeneous environment

Paradox of the plankton

Vol. XCV, No. 882 The American Naturalist May–June, 1961

THE PARADOX OF THE PLANKTON*

G. E. HUTCHINSON

Osborn Zoological Laboratory, New Haven, Connecticut

- Species need different niches to coexist
- How can so many species of plankton coexist?
  - All need the same resources
  - Well-mixed, homogeneous environment

Or, more generally, how can so many species coexist in a given location when they all need more or less the same things (oxygen, water, carbon, etc.)?

How to coexist

We now know the answer (in a broad sense)

What needs to happen for species to coexist?

How to coexist

We now know the answer (in a broad sense)

What needs to happen for species to coexist?

$$\alpha_{ii} > \alpha_{ji}$$

$$\alpha_{jj} > \alpha_{ij}$$

How to coexist

What causes species to limit themselves more than their competitors?

- Different resources
- Different predators

How to coexist

What causes species to limit themselves more than their competitors?

- Different resources
- Different predators
- Spatial variation (use different locations)
- Temporal variation (use different times)
- Some combination of these things

Coexistence via spatial variation

- Variation in resources (or predators) within a patch
- Different species do best with different levels of that resource/predator

Coexistence via spatial variation

- Variation in resources (or predators) within a patch
- Different species do best with different levels of that resource/predators

Requires a trade-off!

Coexistence via spatial variation

- Variation in resources (or predators) within a patch
- Different species do best with different levels of that resource/predators

Requires a trade-off!

Species need to be good at something, bad at others

Coexistence via spatial variation

- Variation in resources (or predators) within a patch
- Different species do best with different levels of that resource/predators

Requires a trade-off!

Species need to be good at something, bad at others

These difference make species more likely to compete with conspecifics

Coexistence via spatial variation

- Variation in resources (or predators) within a patch
- Different species do best with different levels of that resource/predators

Requires a trade-off!

Species need to be good at something, bad at others

These difference make species more likely to compete with conspecifics

Increases intraspecific competition relative to interspecific

Coexistence via spatial variation

- Variation in resources (or predators) within a patch
- Different species do best with different levels of that resource/predators

Requires a trade-off!

Species need to be good at something, bad at others

These difference make species more likely to compete with conspecifics

Increases intraspecific competition relative to interspecific

$$\alpha_{ii} > \alpha_{ji}$$

Trade-offs

- Can be n-dimensional, but we often focus on 2-dimensional trade-offs

Trade-offs

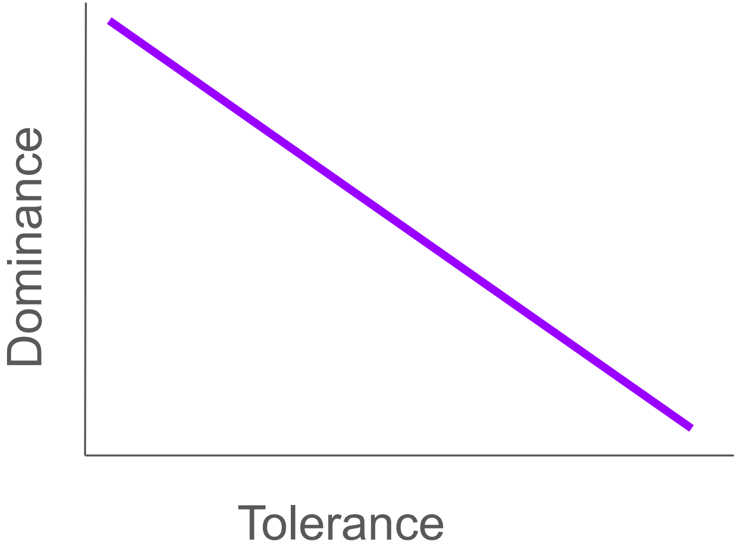
- Can be n-dimensional, but we often focus on 2-dimensional trade-offs

Trade-offs

- Can be n-dimensional, but we often focus on 2-dimensional trade-offs

Trade-offs

- Can be n-dimensional, but we often focus on 2-dimensional trade-offs



Trade-offs

- Can be n-dimensional, but we often focus on 2-dimensional trade-offs

Tolerance-Fecundity

The tolerance–fecundity trade-off and the maintenance of diversity in seed size

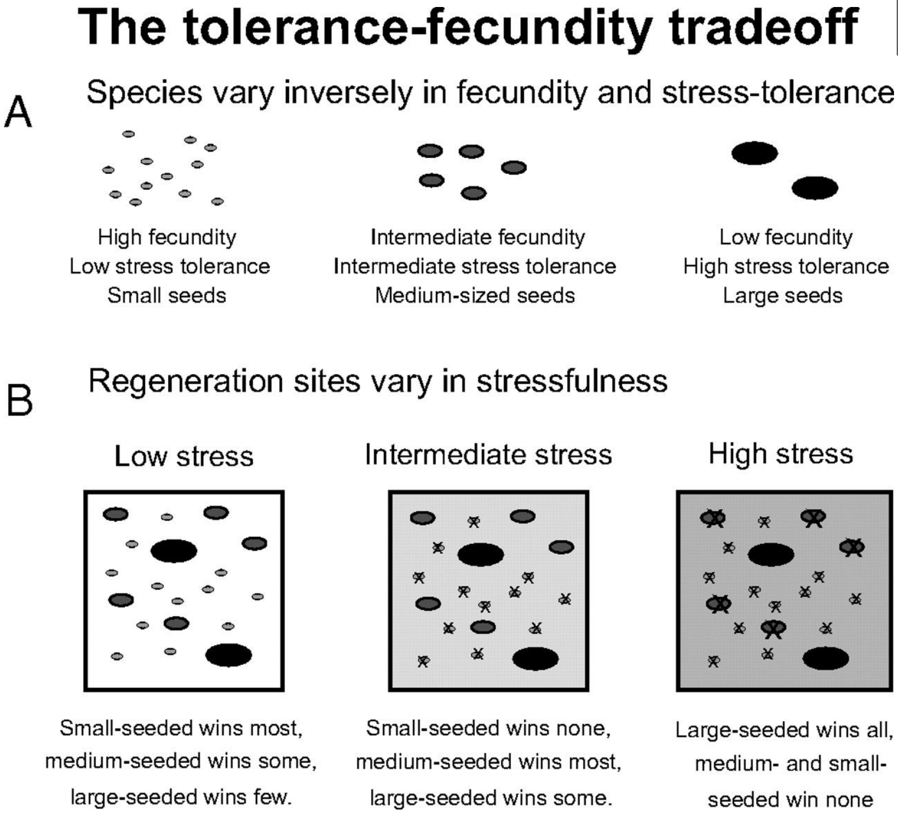
Helene C. Muller-Landau[1]

Smithsonian Tropical Research Institute, DPO AA 34002-9998; and Department of Ecology, Evolution and Behavior, University of Minnesota, St. Paul, MN 55108

Edited* by Simon A. Levin, Princeton University, Princeton, NJ, and approved January 19, 2010 (received for review October 8, 2009)

Seed size commonly varies by five to six orders of magnitude among coexisting plant species, a pattern ecologists have long sought to explain. Because seed size trades off with seed number, small-

nization trade-off (17–19), a mechanism that requires very strong competitive asymmetry–that is, the better competitor must virtually always win no matter how slight its advantage (20). However,



Tolerance-Fecundity


The tolerance–fecundity trade-off and the maintenance of diversity in seed size

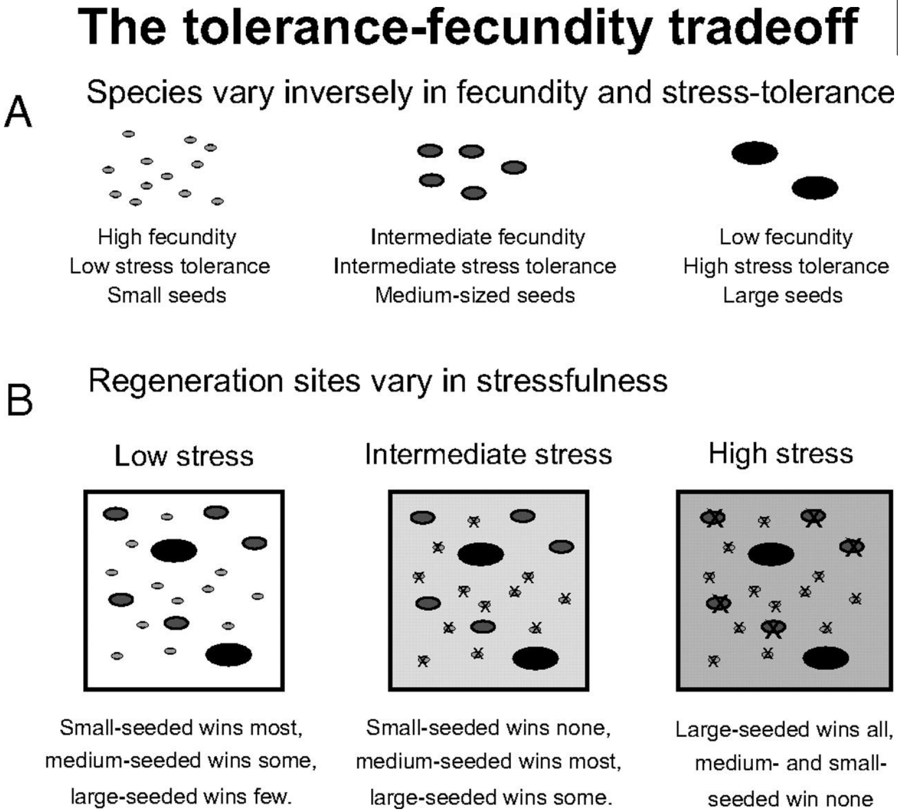
Helene C. Muller-Landau[1]

Smithsonian Tropical Research Institute, DPO AA 34002-9998; and Department of Ecology, Evolution and Behavior, University of Minnesota, St. Paul, MN 55108

Edited* by Simon A. Levin, Princeton University, Princeton, NJ, and approved January 19, 2010 (received for review October 8, 2009)

Seed size commonly varies by five to six orders of magnitude among coexisting plant species, a pattern ecologists have long sought to explain. Because seed size trades off with seed number, small-

nization trade-off (17–19), a mechanism that requires very strong competitive asymmetry–that is, the better competitor must virtually always win no matter how slight its advantage (20). However,



Found theoretical support

Received: 13 March 2018 | Accepted: 23 May 2018
DOI: 10.1111/1365-2745.13028

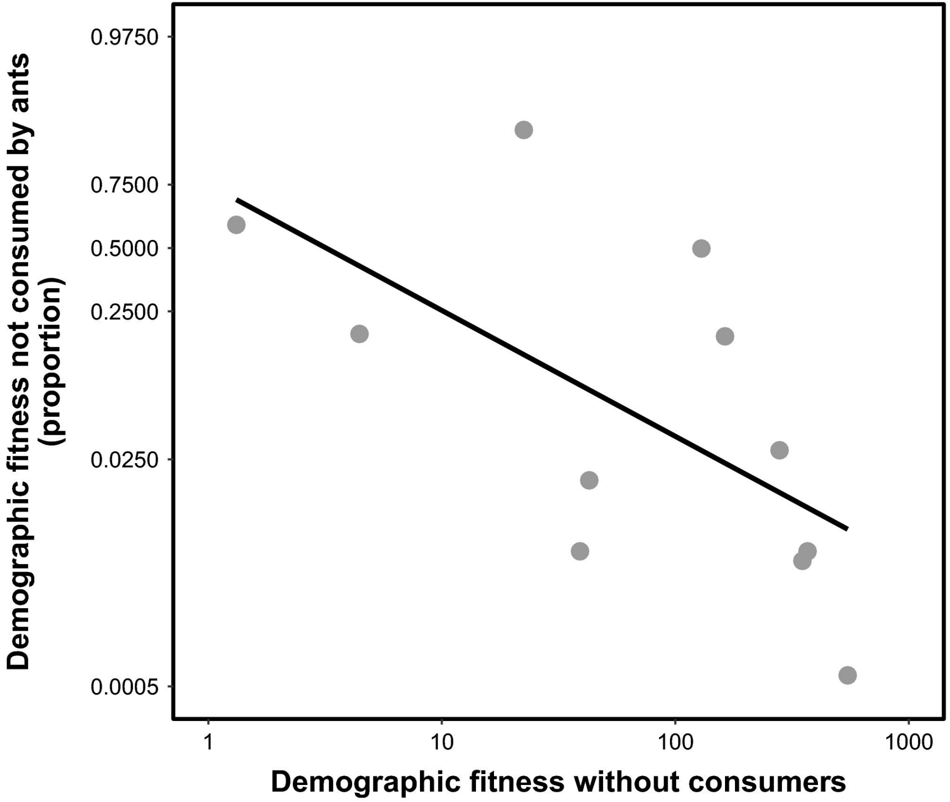
BIOTIC CONTROLS OF PLANT COEXISTENCE

Journal of Ecology BRITISH ECOLOGICAL SOCIETY

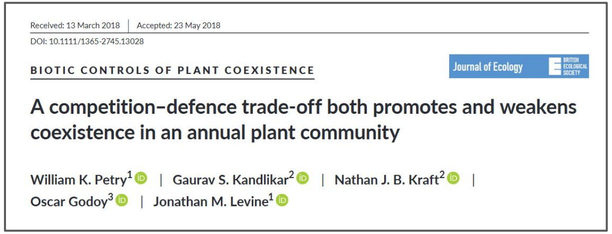
A competition–defence trade-off both promotes and weakens coexistence in an annual plant community

William K. Petry[1] | Gaurav S. Kandlikar[2] | Nathan J. B. Kraft[2] | Oscar Godoy[3] | Jonathan M. Levine[1]

Competition-Defense Trade-off

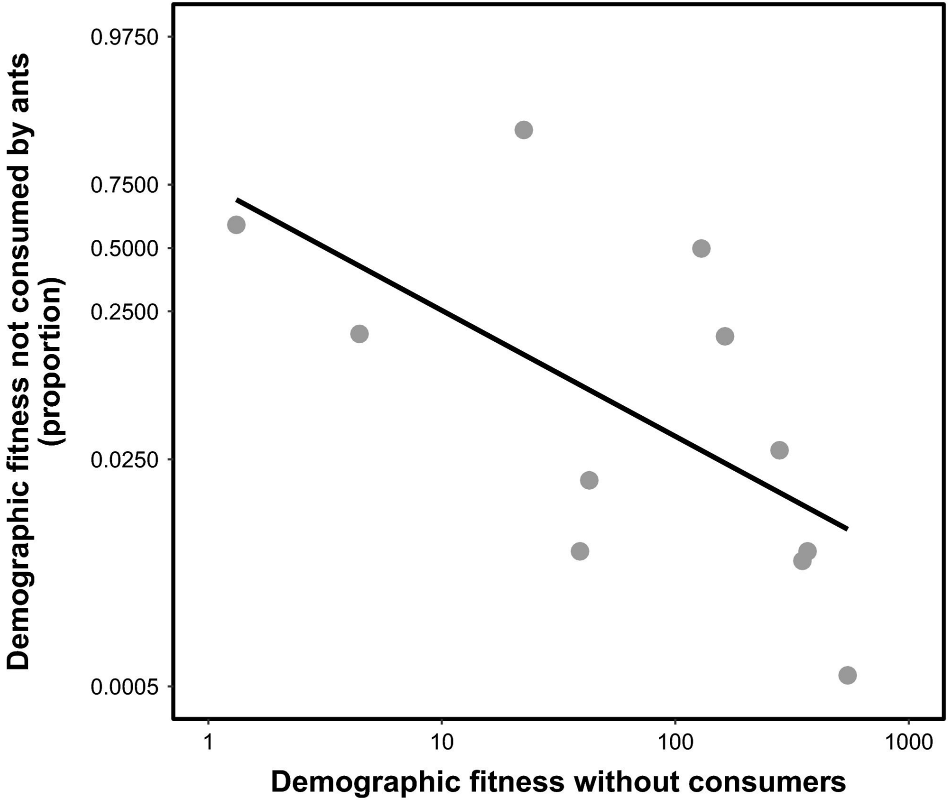


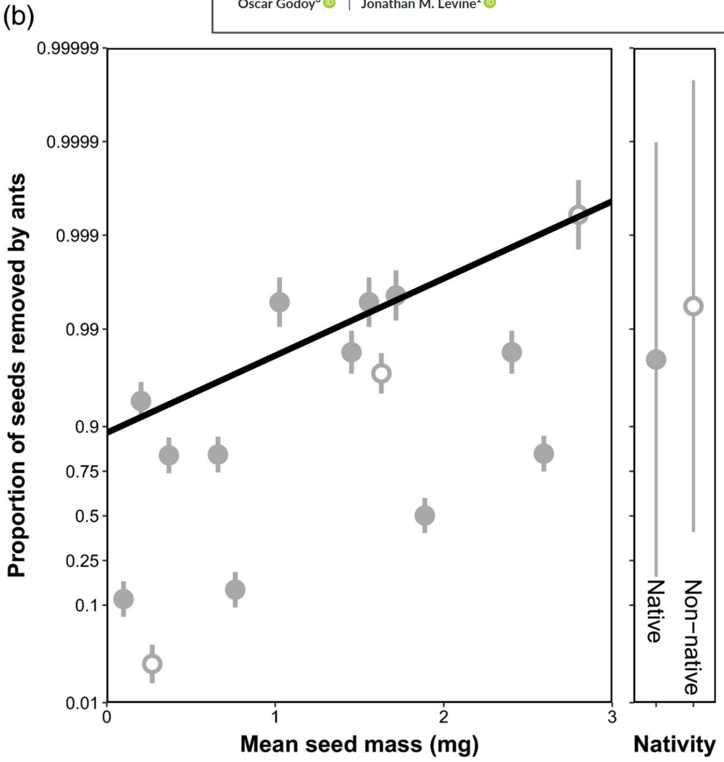
Competition-Defense Trade-off


Received: 13 March 2018 | Accepted: 23 May 2018
DOI: 10.1111/1365-2745.13028
BIOTIC CONTROLS OF PLANT COEXISTENCE
Journal of Ecology

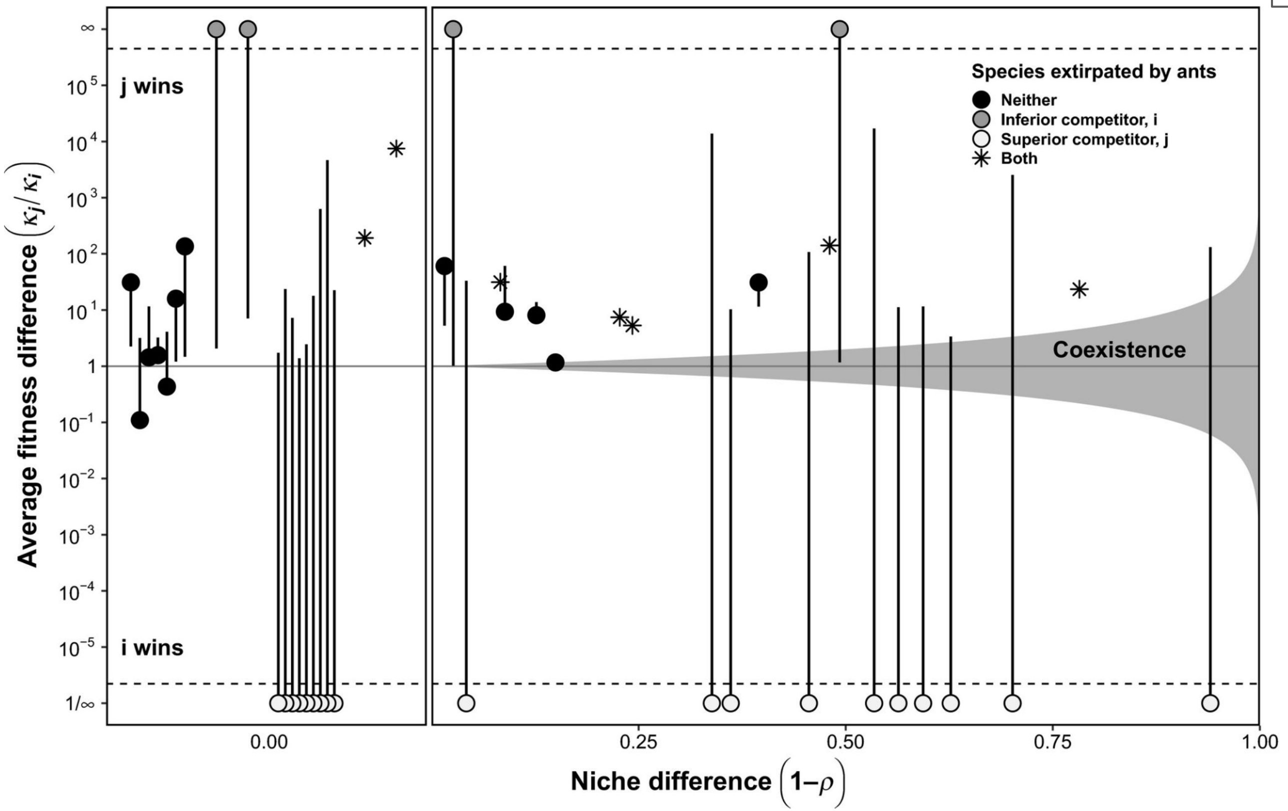
A competition–defence trade-off both promotes and weakens coexistence in an annual plant community

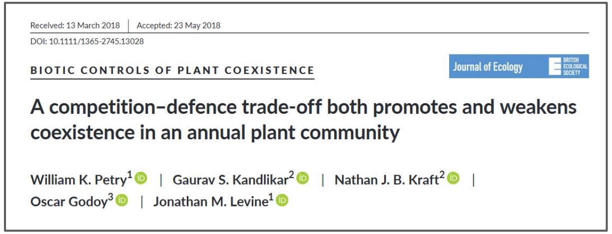
William K. Petry[1] | Gaurav S. Kandlikar[2] | Nathan J. B. Kraft[2] | Oscar Godoy[3] | Jonathan M. Levine[1]





Competition-Defense Trade-off





Competition-Defense Trade-off

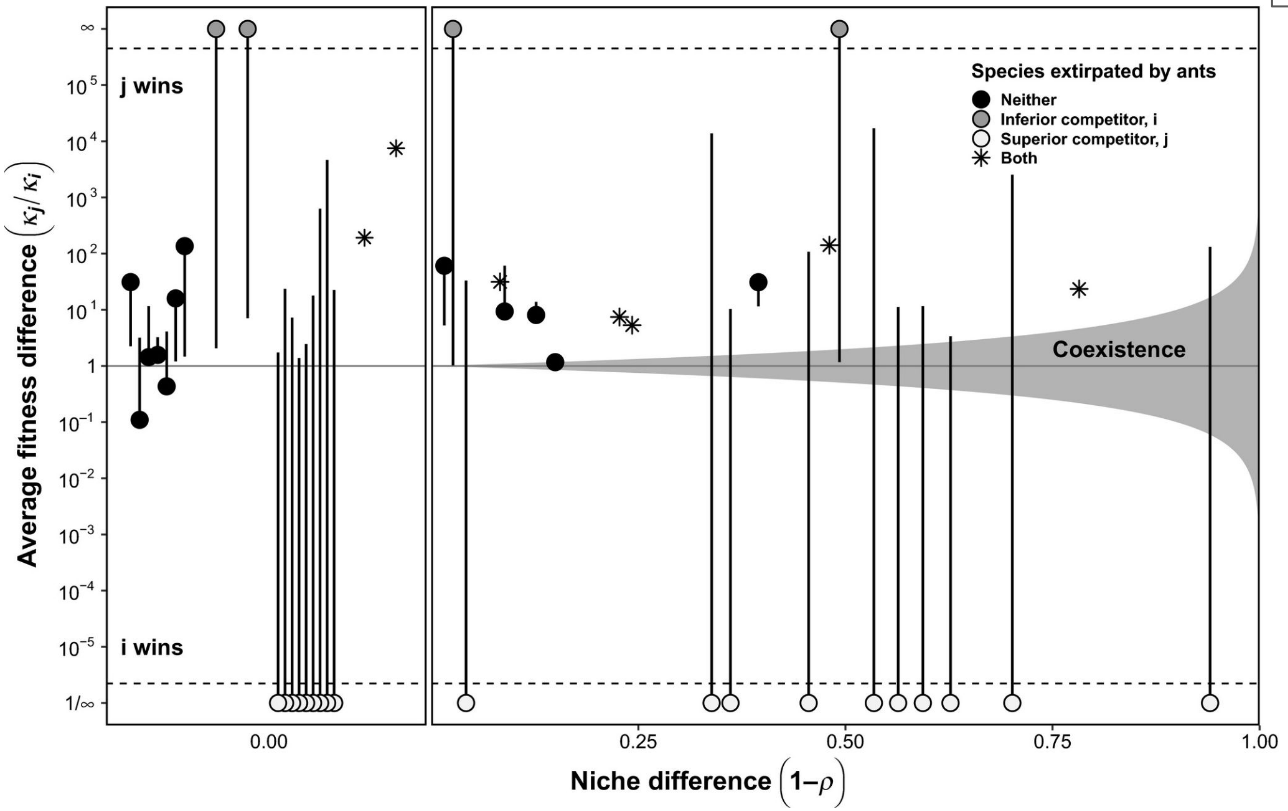
Received: 13 March 2018 | Accepted: 23 May 2018
DOI: 10.1111/1365-2745.13028

BIOTIC CONTROLS OF PLANT COEXISTENCE

Journal of Ecology

A competition–defence trade-off both promotes and weakens coexistence in an annual plant community

William K. Petry[1] | Gaurav S. Kandlikar[2] | Nathan J. B. Kraft[2] | Oscar Godoy[3] | Jonathan M. Levine[1]



Found trade-offs, but…

Trade-off sometimes helped, sometimes made things worse

Coexistence without resource/predation differences

- Different preferences
- Different habitats
- Different strata within a habitat

Coexistence without resource/predation differences

- Different preferences
- Different habitats
- Different strata within a habitat

Same resource, but different locations: increases α_{ii} relative to α_{ji}

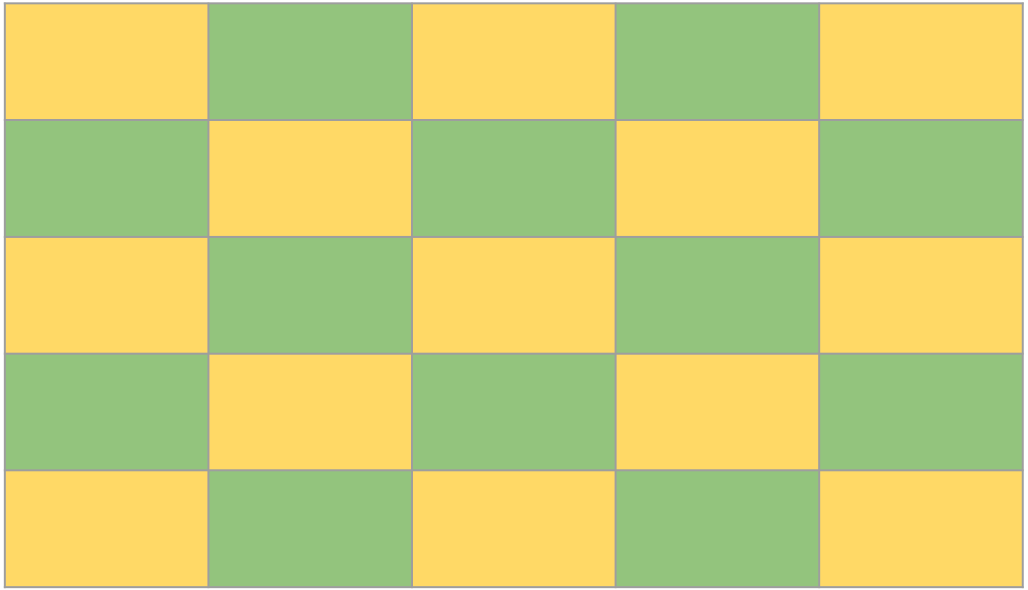
Coexistence without resource/predation differences

- Different preferences
- Different habitats
- Different strata within a habitat

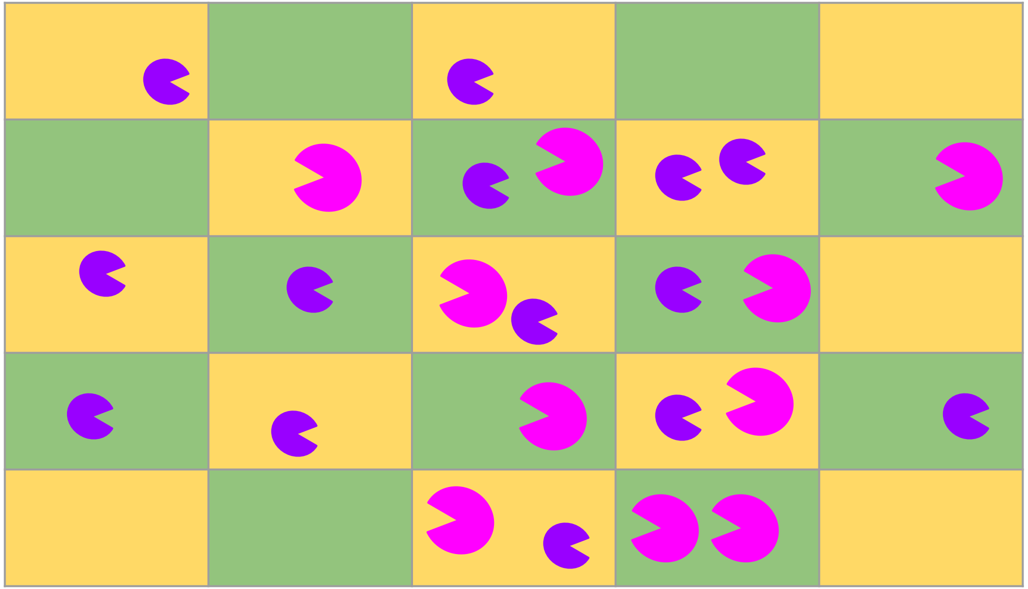
Same resource, but different locations: increases α_{ii} relative to α_{ji}

Note that there are a few caveats about this!

Coexistence without resource/predation differences



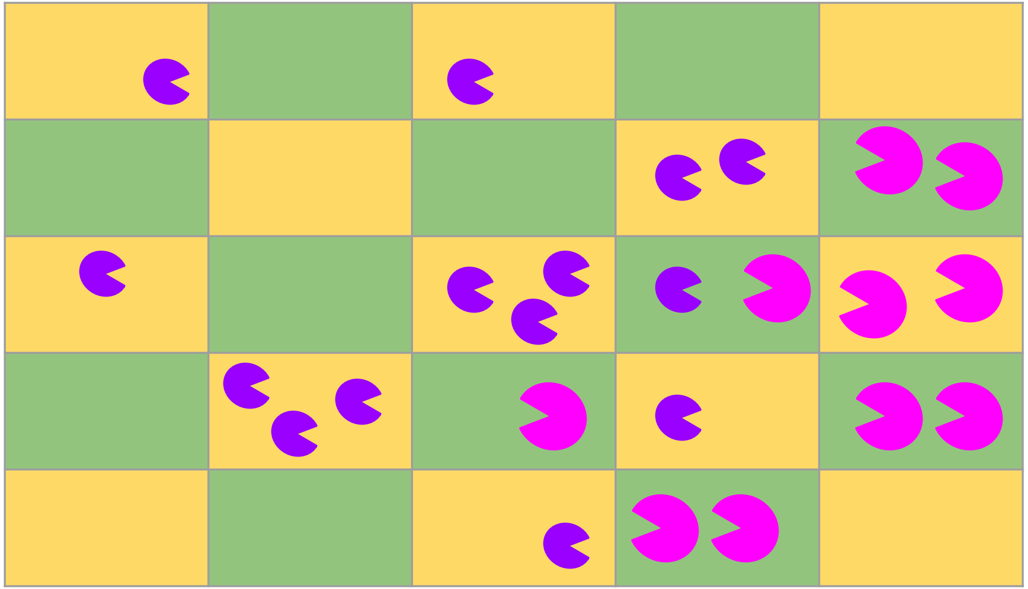
Coexistence without resource/predation differences



= Competitor

Similar preferences

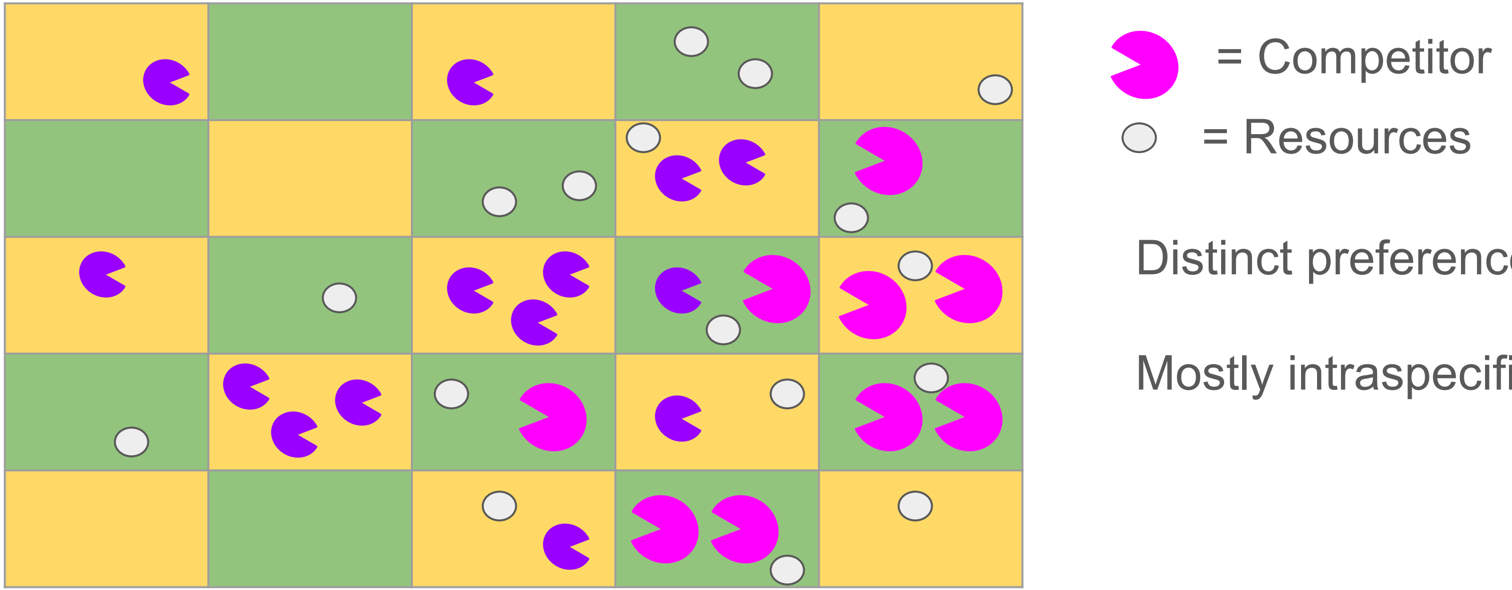
Coexistence without resource/predation differences



= Competitor

Distinct preferences

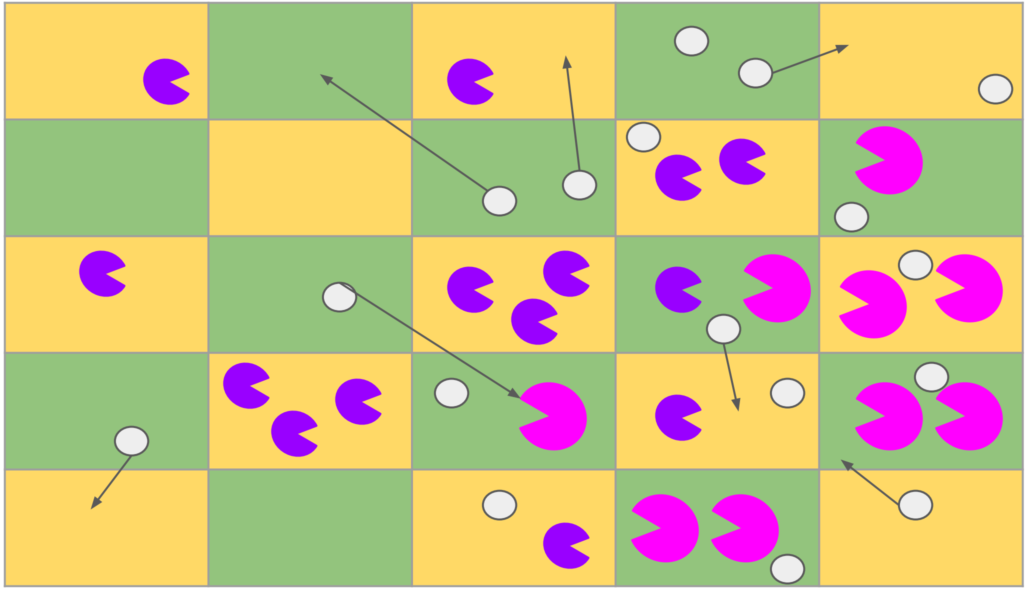
Coexistence without resource/predation differences



Distinct preferences

Mostly intraspecific

Coexistence without resource/predation differences



= Competitor

= Resources
(can now move)

Distinct preferences

No longer consuming unique resource, increases **interspecific** competition

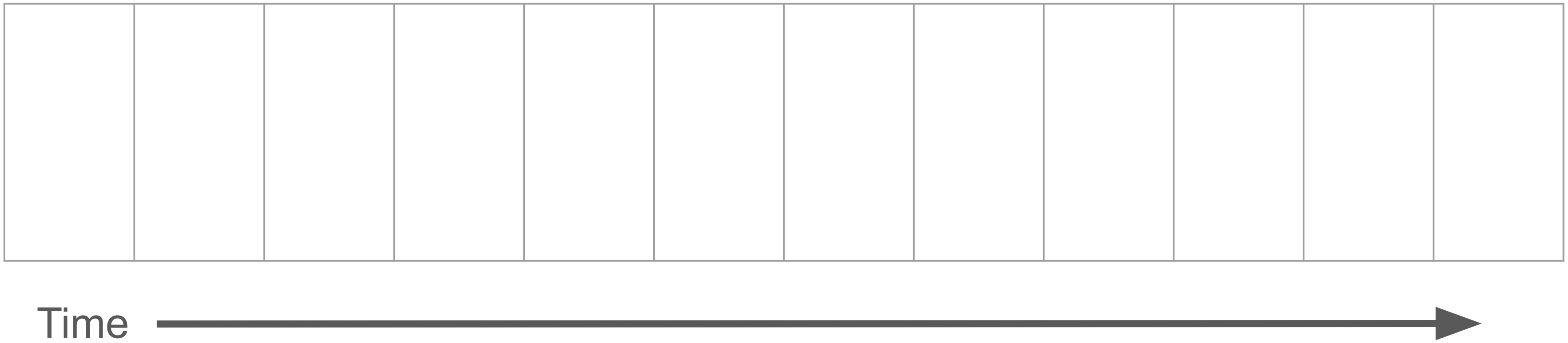
Coexistence via temporal mechanisms

- Temporal niche partitioning
- Relative non-linearity
- Storage effects

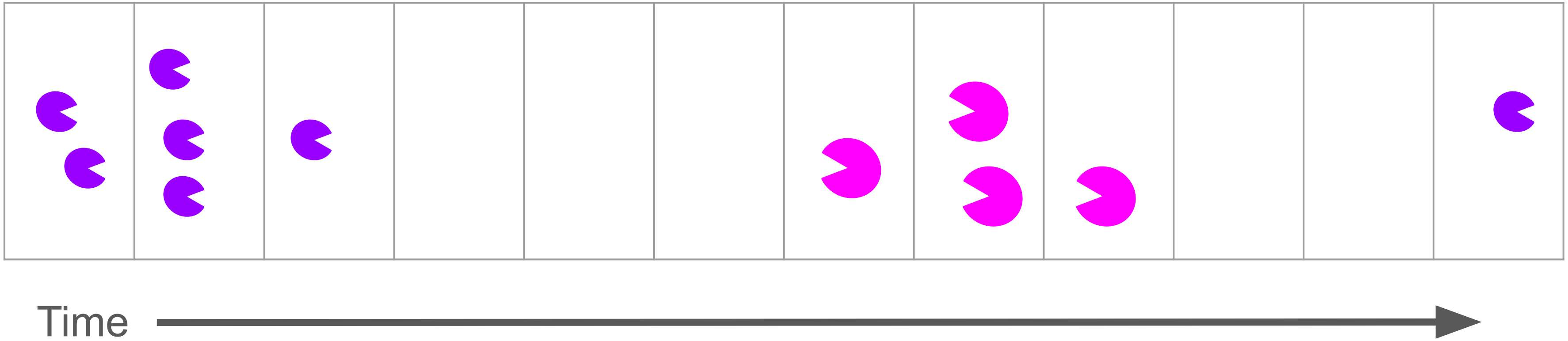
Temporal Niche Partitioning

- Different species use resources at different times
- Note that the resource pools need to be different

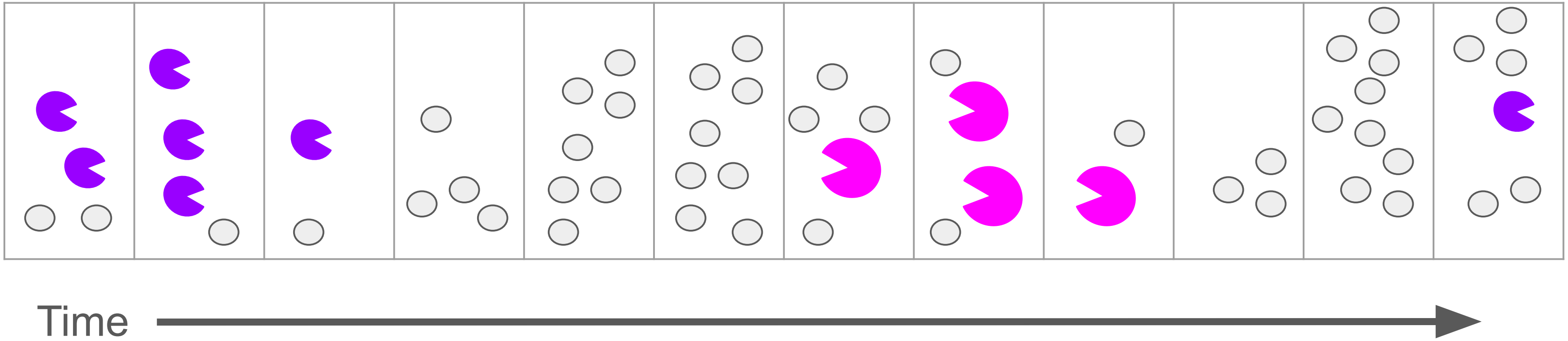
Temporal Niche Partitioning



Temporal Niche Partitioning

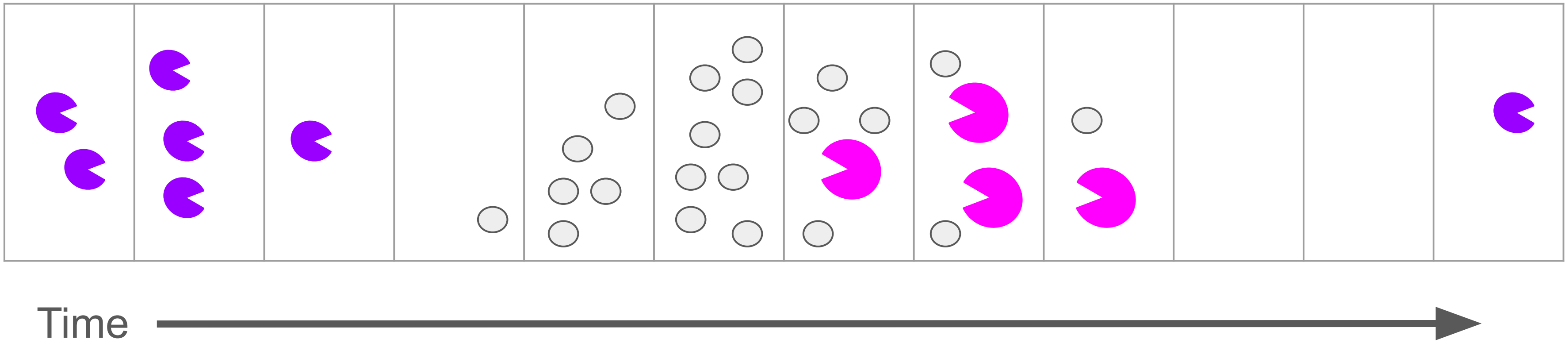


Temporal Niche Partitioning



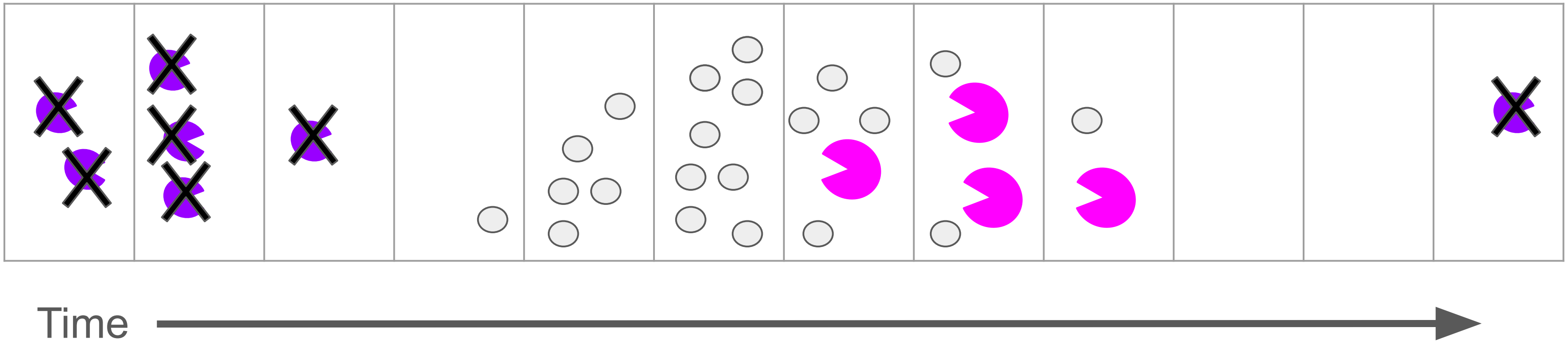
Resources are renewed over time
- can increase intraspecific competition vs interspecific

Temporal Niche Partitioning



Resources are renewed rarely
- dominance by one competitor

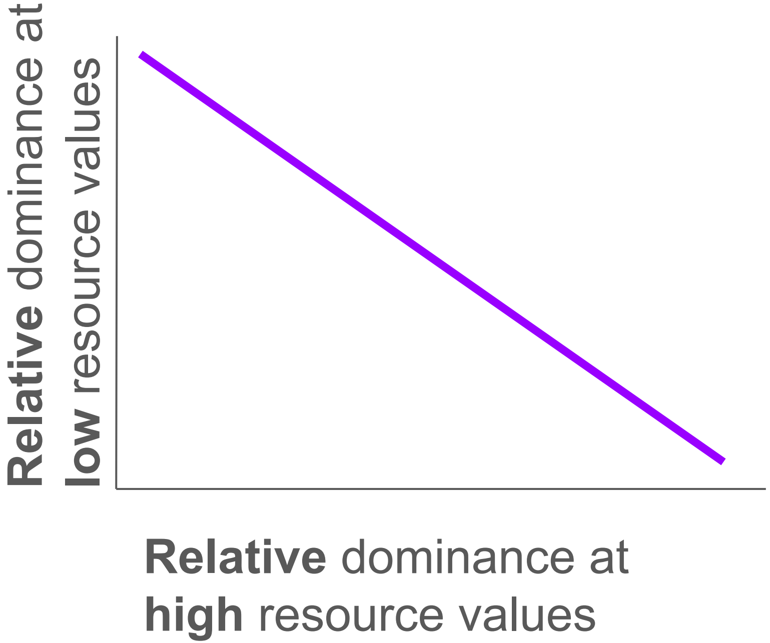
Temporal Niche Partitioning



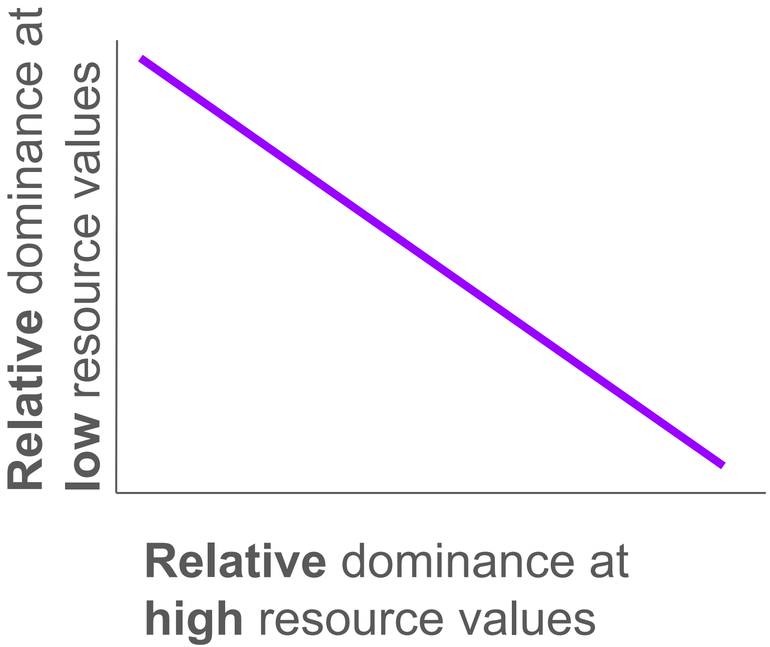
Resources are renewed rarely
- dominance by one competitor

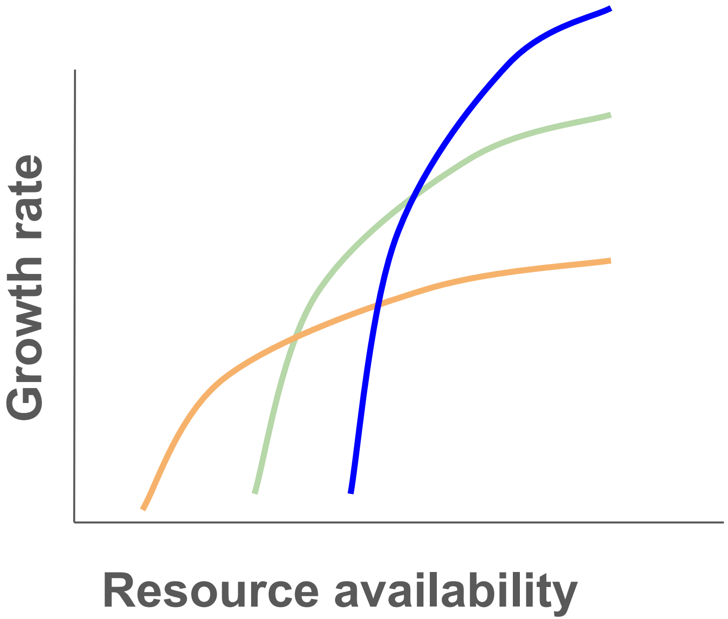
Relative Nonlinearity of competition

Relative Nonlinearity of competition

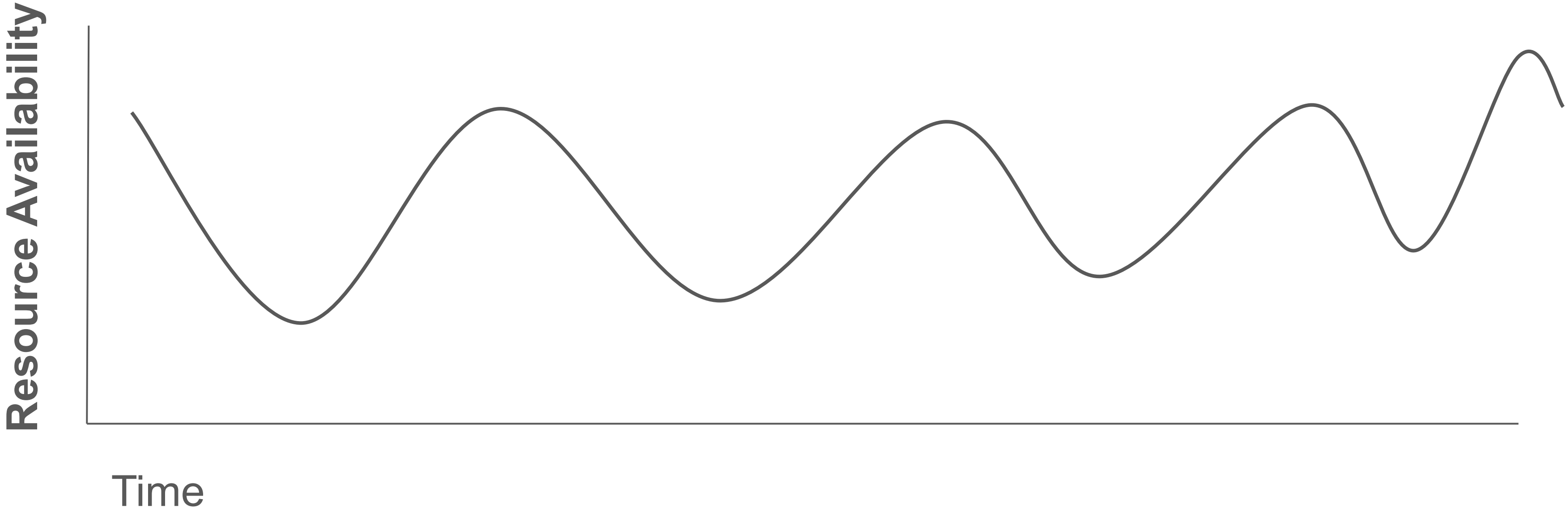


Relative Nonlinearity of competition

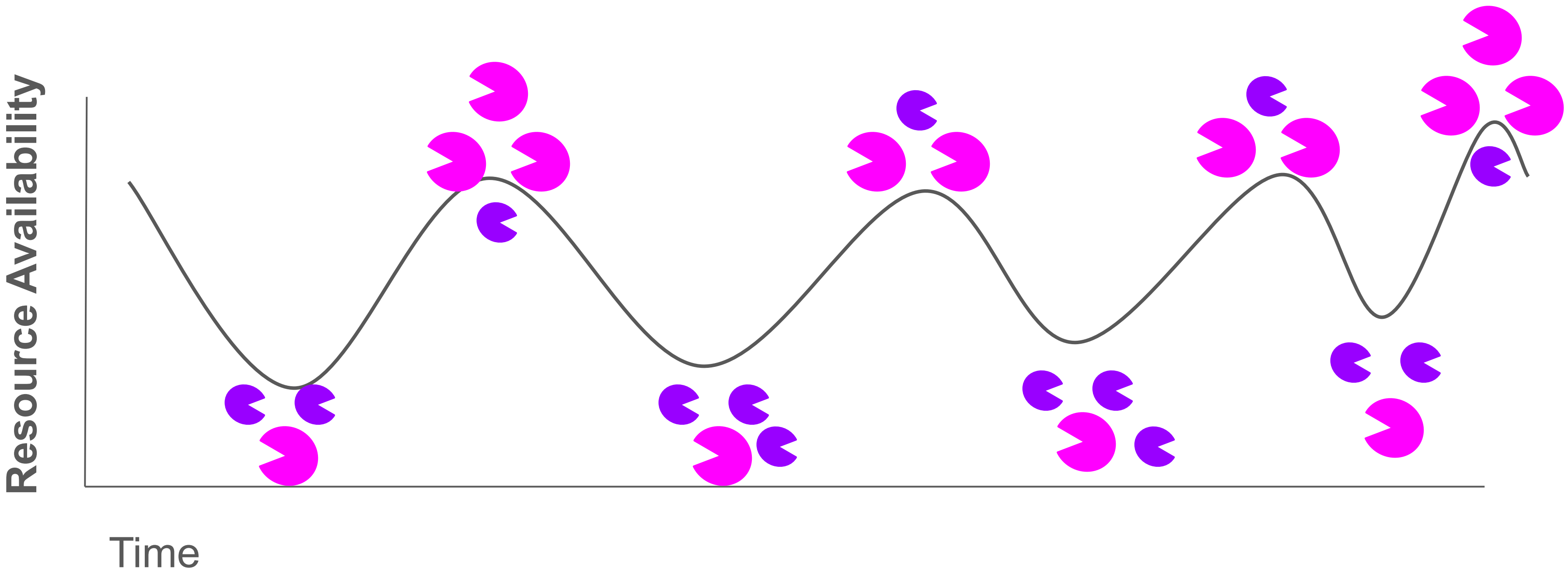




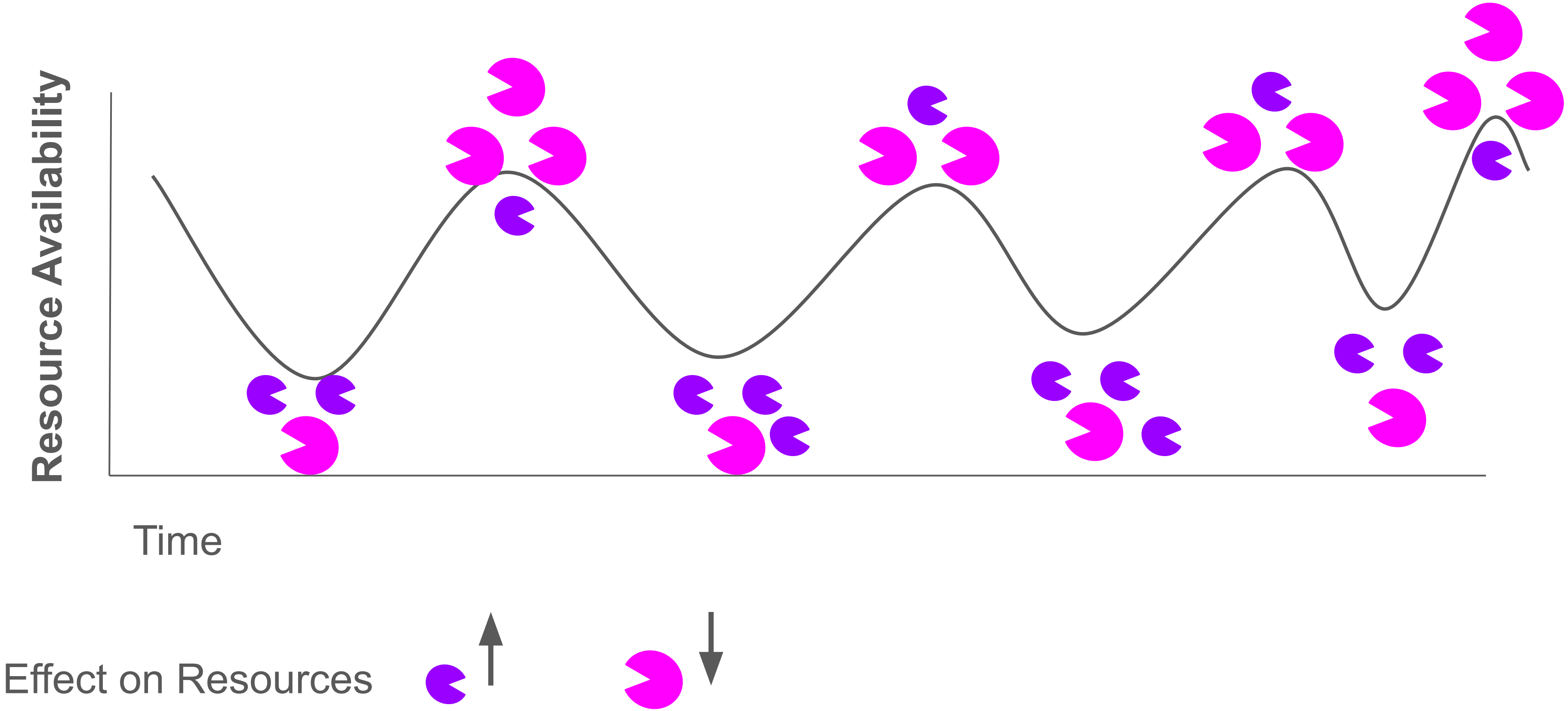
Relative Nonlinearity of competition



Relative Nonlinearity of competition



Relative Nonlinearity of competition

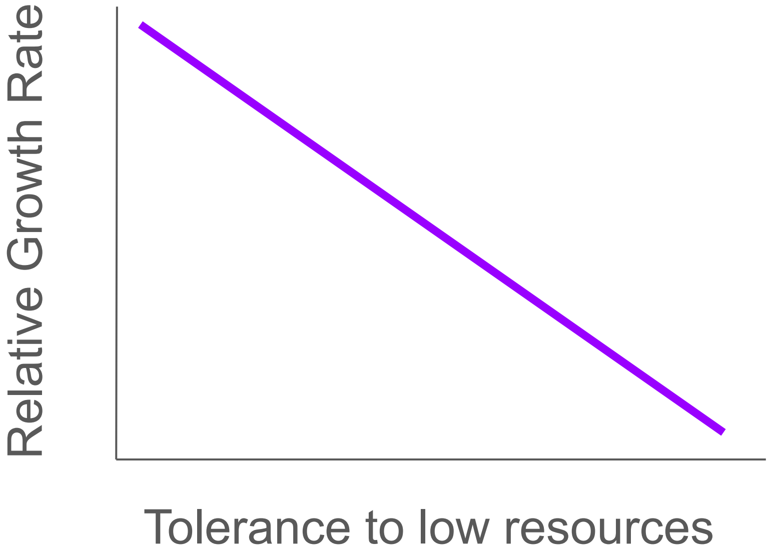


Storage Effect

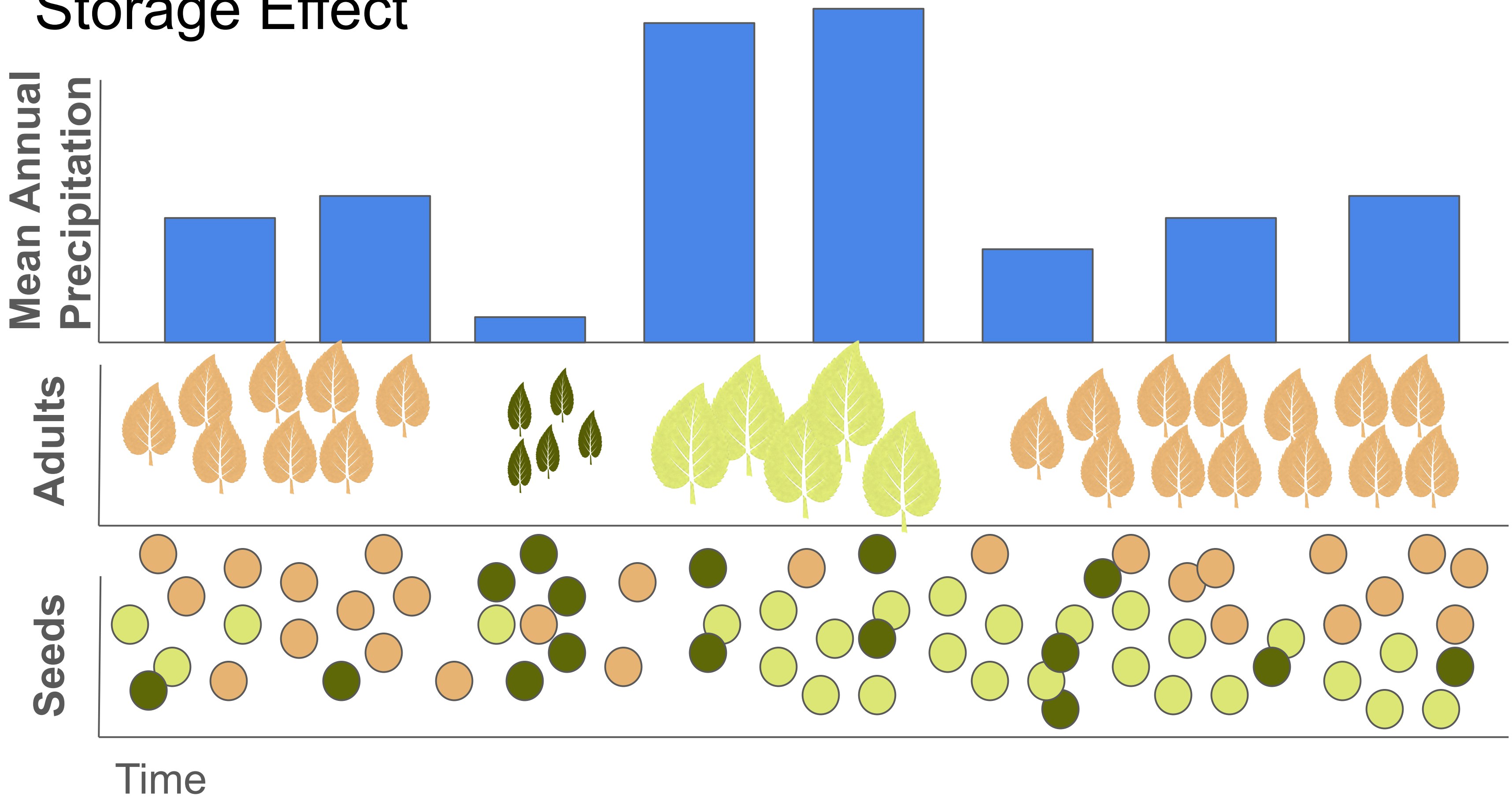
Storage Effect

- Species differ in the environment they do best in
- BUT they can persist through sub-optimal conditions
- (Note that there is a spatial version of this as well)

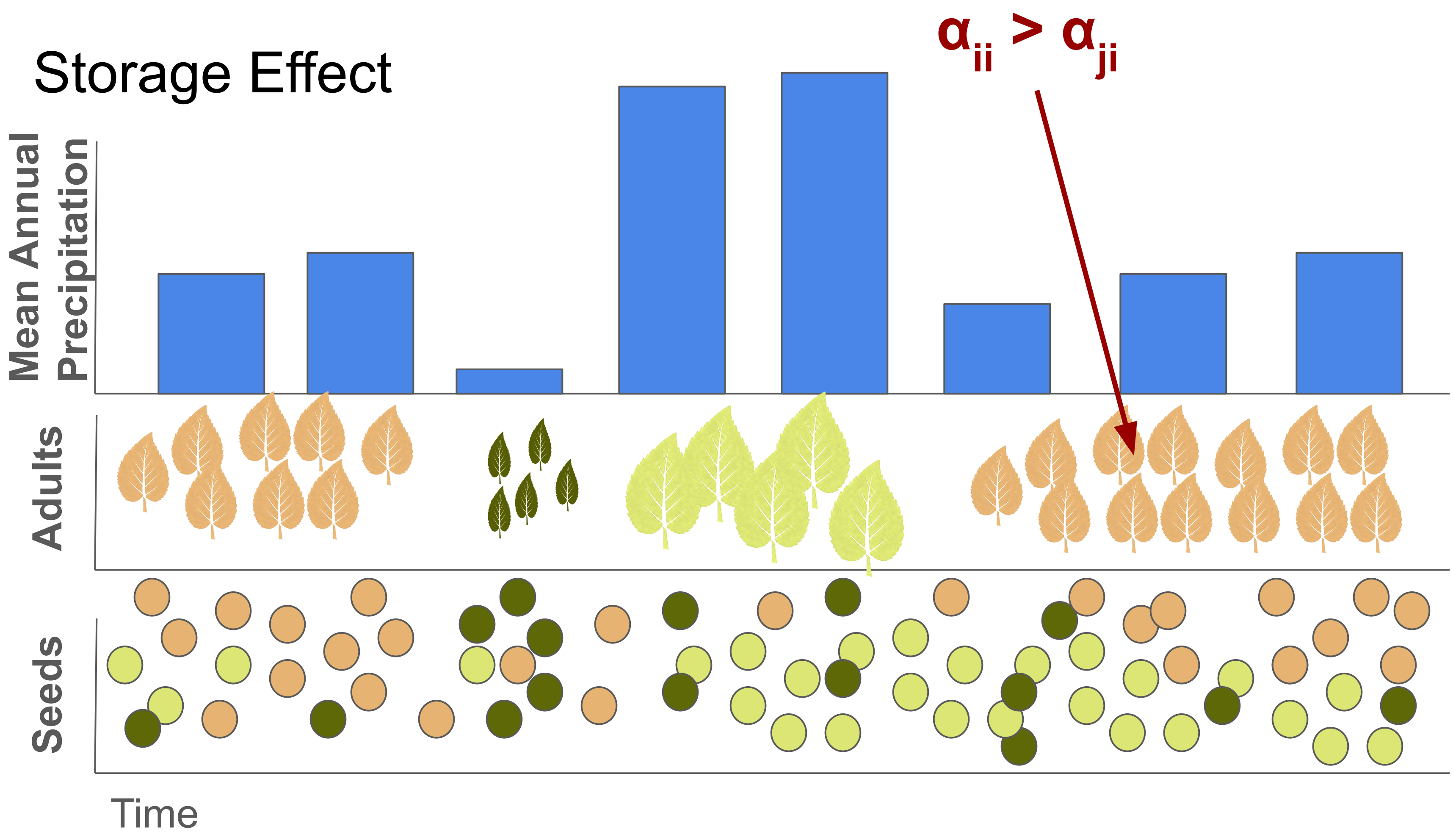
Storage Effect



Storage Effect



Storage Effect



Storage Effect

PNAS

Functional tradeoffs determine species coexistence via the storage effect

Amy L. Angert[a,b,1], Travis E. Huxman[b,c], Peter Chesson[b], and D. Lawrence Venable[b]

[a]Department of Biology, Colorado State University, Fort Collins, CO, 80523; and [b]Department of Ecology and Evolutionary Biology and [c]B2 Earthscience, University of Arizona, Tucson, AZ 85721

Communicated by José Sarukhán, Universidad Nacional Autónoma de México, Mexico D.F., Mexico, April 28, 2009 (received for review March 23, 2009)

How biological diversity is generated and maintained is a fundamental question in ecology. Ecologists have delineated many

demographic decoupling of species arises from partially uncor lated responses to environmental variation, (*ii*) the strength

Storage Effect

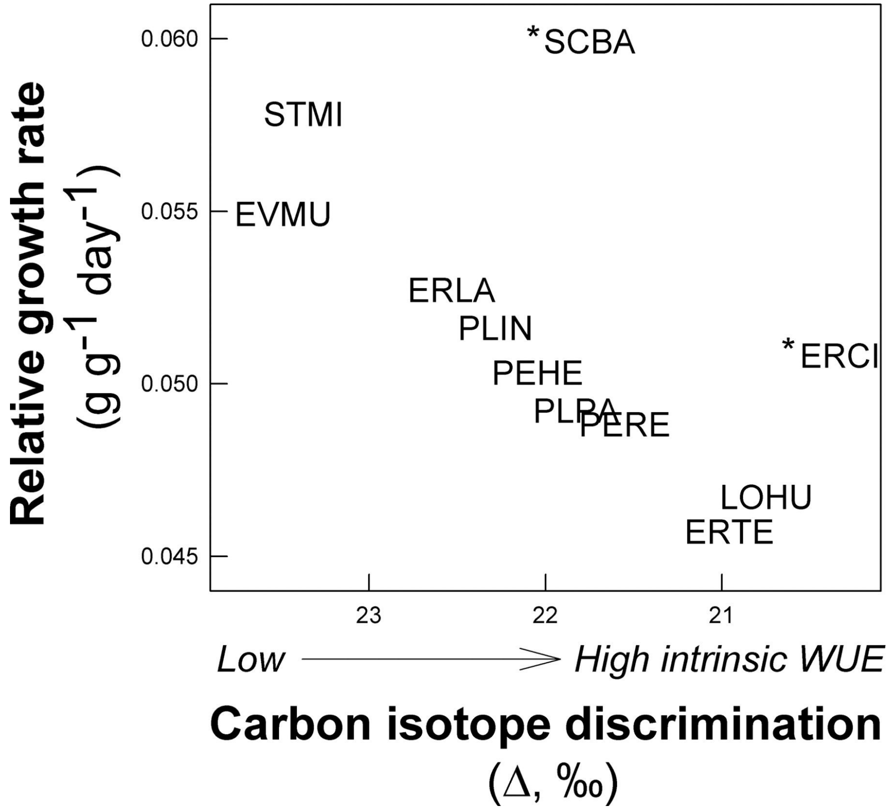
Functional tradeoffs determine species coexistence via the storage effect

Amy L. Angert[a,b,1], Travis E. Huxman[b,c], Peter Chesson[b], and D. Lawrence Venable[b]

[a]Department of Biology, Colorado State University, Fort Collins, CO, 80523; and [b]Department of Ecology and Evolutionary Biology and [c]B2 Earthscience, University of Arizona, Tucson, AZ 85721

Communicated by José Sarukhán, Universidad Nacional Autónoma de México, Mexico D.F., Mexico, April 28, 2009 (received for review March 23, 2009)

How biological diversity is generated and maintained is a fundamental question in ecology. Ecologists have delineated many demographic decoupling of species arises from partially uncor lated responses to environmental variation, (*ii*) the strength



Storage Effect


PNAS

Functional tradeoffs determine species coexistence via the storage effect

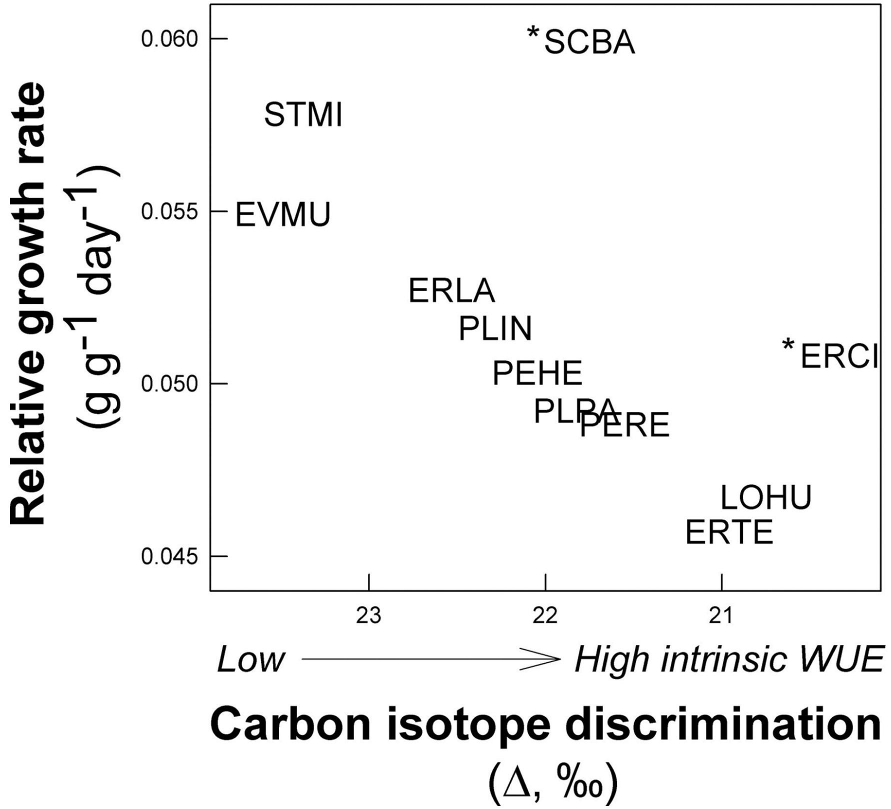
Amy L. Angert[a,b,1], Travis E. Huxman[b,c], Peter Chesson[b], and D. Lawrence Venable[b]

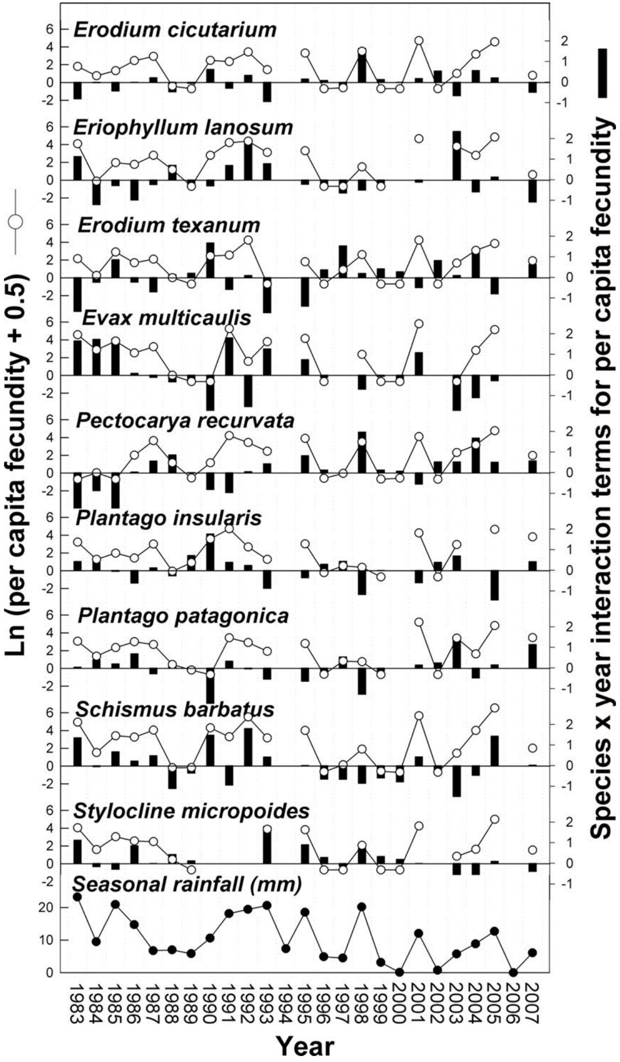
[a]Department of Biology, Colorado State University, Fort Collins, CO, 80523; and [b]Department of Ecology and Evolutionary Biology and [c]B2 Earthscience, University of Arizona, Tucson, AZ 85721

Communicated by José Sarukhán, Universidad Nacional Autónoma de México, Mexico D.F., Mexico, April 28, 2009 (received for review March 23, 2009)

How biological diversity is generated and maintained is a fundamental question in ecology. Ecologists have delineated many

demographic decoupling of species arises from partially uncor lated responses to environmental variation, (ii) the strength





Storage Effect

Functional tradeoffs determine species coexistence via the storage effect

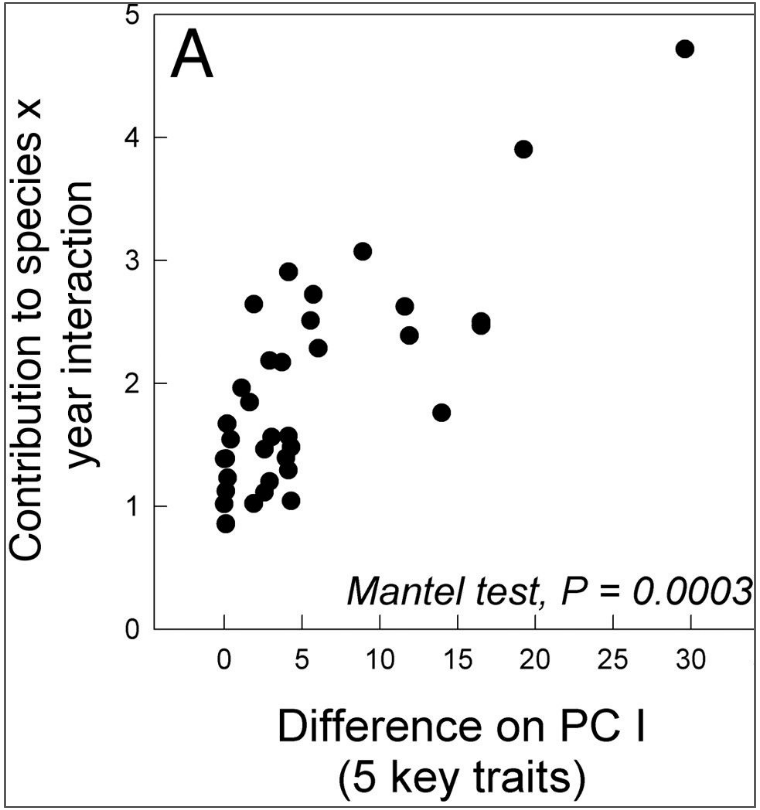
Amy L. Angert[a,b,1], Travis E. Huxman[b,c], Peter Chesson[b], and D. Lawrence Venable[b]

[a]Department of Biology, Colorado State University, Fort Collins, CO, 80523; and [b]Department of Ecology and Evolutionary Biology and [c]B2 Earthscience, University of Arizona, Tucson, AZ 85721

Communicated by José Sarukhán, Universidad Nacional Autónoma de México, Mexico D.F., Mexico, April 28, 2009 (received for review March 23, 2009)

How biological diversity is generated and maintained is a fundamental question in ecology. Ecologists have delineated many

demographic decoupling of species arises from partially uncor lated responses to environmental variation, (*ii*) the strength



Main Takeaways

- Species coexistence requires meaningful differences
- Species can differ in resource/predation in space and/or time
- Trade-offs are critical!
- Trade-offs need to make $\boldsymbol{\alpha_{ii} > \alpha_{ji}}$ for coexistence

Next class: Mutualism and Facilitation